HOMEWORK 3.8 · UNIT 3 · NC.8.EE.7

Inequalities (Day 2) & Number of Solutions (Day 2)

Targets: EE.7-4, EE.7-7, EE.7-3, EE.7-6, EE.7-8, EE.7-9

Name _____ Date _____ Block _____

Remember: How many solutions? Simplify both sides. Different x-coefficients → **one**. Same x-coefficient, different numbers → **none**. Both sides exactly the same → **infinitely many**.
Example: $2(x+3) = 2x+6 \to 2x+6 = 2x+6 \to$ infinitely many. Inequalities: flip only for × or ÷ by a negative.

SCORE 3.0 Approaching · the basics

Targets: EE.7-4, EE.7-7, EE.7-3

1 How many solutions? $4x + 9 = 4x + 9$ EE.7-4

One solution | No solution | Infinitely many

2 How many solutions? $2(x + 1) = 2x + 5$ EE.7-4

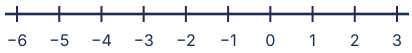
One solution | No solution | Infinitely many

3 Which inequality matches the graph? REVIEW 3.7 EE.7-7



Ⓐ $x < 1$ Ⓑ $x \le 1$ Ⓒ $x > 1$ Ⓓ $x \ge 1$

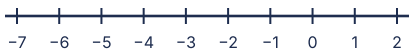
4 Solve and graph. $x + 3 \ge 1$ REVIEW 3.7 EE.7-7



5 Solve. $-2(x + 1) = x - 11$ REVIEW 3.6 EE.7-3

SCORE 4.0 Proficient · on target Targets: EE.7-7, EE.7-6

6 Solve and graph. $-4x - 3 > 9$ EE.7-7



7 How many solutions does $7x - 3(x + 2) = 4x - 6$ have? EE.7-6

(A) None (B) One
(C) Infinitely many (D) One: $x = 0$

8 **Zoe** made a mistake. Circle it. Explain it. Fix it. EE.7-7

ZOE'S WORK

$2x - 5 \geq 4x + 3$

$-2x - 5 \geq 3$

$-2x \geq 8$

$x \geq -4$

SCORE 5.0 Mastery · go deeper Targets: EE.7-8, EE.7-9

9 Change **one** number in $2(x + 4) = 2x + 8$ so it has **no solution**. Explain. EE.7-8

My new equation: _____ . It has no solution because _____ .

10 Marco has \$28. He earns \$9 an hour. He wants **at least** \$100. Solve $28 + 9h \geq 100$. What does it mean? EE.7-9

h _____ . This means _____ .